PRD v2.0 - Week 13 Update
Maui Luxury Vacation Rentals Dashboard
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SECTION 1: PROPERTY AND AUDIENCE

Property Name: Maui Luxury Vacation Rentals Dashboard
Island: Maui
Property Type: Vacation Rental (Airbnb-style)
Target Segment: Canadian travelers escaping winter

This project is a React application for a small group of Maui vacation rentals. The primary audience is Canadian couples and small families from cities such as Vancouver, Calgary, and Toronto who usually book 5-10 night trips between November and March. They compare value, comfort, and location before booking. Compared to v1.0, the audience is unchanged. v2.0 expands the product with a statistics dashboard, island filtering, and weather context so both guests and the property operator can make faster decisions.

SECTION 2: PROBLEM STATEMENT

The app solves two connected problems.

Operator problem: Kai manages five properties but currently compares pricing, stay duration, and demand manually. He needs one dashboard view to support pricing and minimum-night decisions during peak Canadian season.

Visitor problem: A Canadian traveler (for example, Claire from Vancouver) needs to compare listings quickly and understand likely weather conditions before booking. A marketing page alone explains the brand but does not provide side-by-side comparison or simple data visuals.

v2.0 addresses this by pairing the existing marketing page with a data-driven dashboard focused on rates, stay behavior, seasonality, and weather.

SECTION 3: MARKETING PAGE STATUS (FROM v1.0)

The marketing page is complete in week12/term-project and remains stable in v2.0. It uses six React components: Header, HeroSection, About, Amenities, CTASection, and Footer.

Current behavior:
- Hero presents property identity and positioning.
- About explains value for Canadian guests.
- Amenities renders cards with map().
- CTA includes Contact Us and View More Listings actions.

Technical notes:
- Layout uses CSS Grid/Flexbox with mobile-first responsiveness.
- Percent-based sizing and clamp() improve readability.
- Content is readable at 375px and desktop widths.
- CTA dashboard link is still a placeholder and will be routed in Week 14.

SECTION 4: DASHBOARD DESIGN

The Visitor Statistics Dashboard uses three Chart.js visualizations.

Wireframe (simplified):

```
+---------------------------+  +------------------------------+
| Bar: Nightly Rate         |  | H-Bar: Avg Stay by Segment   |
+---------------------------+  +------------------------------+
+-------------------------------------------------------------+
| Line: Monthly Arrivals (Canada vs Total)                    |
+-------------------------------------------------------------+
```

Chart 1: Nightly Rate by Property (Bar)
- Data source: MongoDB properties collection via GET /api/properties
- Key field: nightlyRate
- Purpose: lets guests compare value quickly before opening detail pages.

Chart 2: Average Stay by Visitor Segment (Horizontal Bar)
- Data source: DBEDT visitor statistics
- Key fields: VisitorOrigin, AverageNightsStayed
- Purpose: shows whether Canadian travelers stay longer, supporting policy choices such as minimum-night settings.

Chart 3: Monthly Arrivals (Line)

- Data source: DBEDT monthly arrivals
- Key fields: Month, TotalArrivals, CanadaArrivals
- Purpose: exposes seasonality, especially winter demand spikes that justify targeted marketing and dynamic pricing.

SECTION 5: ISLAND SELECTOR UX

Control: dropdown select.

Why dropdown:
- Works reliably on mobile where tabs can overflow.
- Reuses interaction pattern from the Week 13 island assignment.
- Avoids extra dependency complexity from map-based selection.

State ownership:
- selectedIsland is managed in DashboardPage via useState.
- DashboardPage is the common parent for all components that consume this state.

Re-render behavior on island change:
- WeatherWidget updates query location.
- ChartSection re-filters chart data.
- PropertyGrid updates visible property cards.

SECTION 6: WEATHER WIDGET

Placement: top-right of the dashboard header, next to island selector.
This keeps weather context tied to the selected island.

API target: Kihei, HI, US (OpenWeatherMap query q=Kihei,HI,US).
Kihei is used because the flagship listing is Kihei Studio by the Sea and it is representative of South Maui visitor conditions.

Displayed fields:
- Temperature in F and C
- Condition text
- Weather icon
- Humidity
- Last-updated timestamp

Resilience behavior:

- Loading: show "Loading weather..."
- Error: show "Weather data unavailable. Please try again later."
- Failure is isolated so the rest of the dashboard still functions.

SECTION 7: DATA ARCHITECTURE

Primary external source:
- Hawaii DBEDT visitor research data (CSV)

Fields used:
- VisitorOrigin
- IslandVisited
- AverageNightsStayed
- TotalArrivals
- Month, Year

Pipeline:
1. CSV is cleaned and standardized.
2. scripts/seedStats.js converts rows to JSON objects.
3. Records are inserted into MongoDB collection visitorStats.
4. Express route GET /api/stats/visitors serves filtered results.

Example schema:
```
{
  "month": "January",
  "year": 2024,
  "island": "Maui",
  "origin": "Canada",
  "totalArrivals": 12400,
  "avgNightsStayed": 7.2
}
```

API filtering:
- /api/stats/visitors?island=Maui&origin=Canada&year=2024

Current status:
- Week 13 charts use local mock data for structure and styling.
- Week 14 replaces mock data with useEffect + fetch integration.

SECTION 8: WEEK 14-15 PLAN

Planned features:
1. Connect charts and cards to live API responses.
2. Add React Router from marketing page CTA to DashboardPage.
3. Add AdminDashboard route with JWT-based authentication.
4. Add property detail page with larger gallery and inquiry form.

Open decisions:
- Weather refresh strategy: periodic refresh vs on-change only.
- Data freshness strategy: direct fetch each load vs server-side TTL cache.

COMPONENT HIERARCHY v2.0

```
App
|- MarketingPage
|  |- Header
|  |- HeroSection
|  |- About
|  |- Amenities
|  |- CTASection
|  |- Footer
|- DashboardPage
|  |- DashboardHeader
|  |  |- IslandSelector
|  |  |- WeatherWidget
|  |- ChartSection
|  |  |- BarChart
|  |  |- HBarChart
|  |  |- LineChart
|  |- PropertyGrid
|     |- PropertyCard
|- AdminDashboard (planned)
   |- AdminPanel
```

ACCEPTANCE CRITERIA

1. Given a visitor opens the marketing page at 375px width, when the page loads, then Hero, About, Amenities, and CTA are readable with no horizontal scroll.

2. Given DashboardPage loads with default island Maui, when weather resolves, then widget shows Kihei temperature in both F and C within 3 seconds under normal network conditions.

3. Given property data is loaded, when the nightly-rate bar chart renders, then highest and lowest priced listings are visually distinguishable.

4. Given arrivals data is loaded, when the line chart renders, then months Jan-Dec appear on x-axis and Canada vs Total series are clearly distinct.

5. Given selectedIsland changes, when user picks a new island, then charts, property grid, and weather widget update without full page reload.

6. Given a visitor clicks Contact Us in CTA, when mail client opens, then the destination address is prefilled and the link does not error.